

Milan George

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Profile

Gameplay and systems programmer specializing in C++ and Unreal Engine 5, with four years building scalable gameplay systems, navigation, tools and performance critical simulation. Project lead on a Total War style RTS, where the battle layer runs on Mass Entity/ECS and sustains 4,000 individually simulated soldiers at 60 FPS.

Skills

Programming: C++ (11 and 17), C#, C, HLSL | **Languages:** English, Malayalam, Hindi

Game Development: Unreal Engine 5, Unity, Gameplay Systems, Blueprints, Mass Entity, ECS, World Partition, Niagara, FMOD

AI & Navigation: A* Grid Pathfinding, Formation Movement, Squad Ability Frameworks, Behaviour Arbitration, State Machines

Performance & Tools: Unreal Insights, CPU/GPU/Memory Profiling, Instanced Rendering, Per Instance LOD Tiering, Object Pooling

Tools & UI: In Editor Authoring Tools, Blueprint Exposed C++ APIs, MVVM Data Binding, UMG, Slate, Common UI

Architecture: Data Oriented Design, ECS, Event Driven Systems, Multithreading, Plugin and Module Architecture, Vertex Animation

Other: Windows, Linux, REST APIs, UnrealBuildTool, CMake, Git, Plastic SCM, Visual Studio, Rider

Experience

Project Lead, Nytwolf Games, Bangalore, India

April 2024 to Present

- Led the development of the studio's custom **RTS battle plugin**; architected **Mass Entity (ECS)** processors for squad formation, movement and combat, converting per actor logic into frame amortized batch simulation for **4,000 units at 60 FPS**.
- Paired every simulation entity with a light presentation actor rendering through **vertex animation**, and built **per instance LOD tiering** and actor pooling to keep crowd rendering affordable at that unit count.
- Built the plugin's profiling and debugging tools: a CSV and **Unreal Insights** timing macro applied across the battle systems, a schema versioned JSONL logger for AI decisions, and console diagnostics for live inspection. Used them to find a **World Partition** leak that kept battle cell worlds alive after the return to the campaign map.
- Engineered a **squad ability framework** covering **18 combat and traversal abilities** including melee, ranged, flank, wall climb, gate breach and siege tower boarding, with priority based arbitration so a player order takes over from the squad's own decision.
- Designed and implemented the **A* grid navigation** and pathing layer in **C++**, with a pivot based movement system that holds squad formation through corners.
- Built the per soldier **animation state machine** on plain C++ state objects instead of **UObjects**, pooled per soldier and reused across battles, so a state change allocates nothing and adds no garbage collection work.
- Architected the project's dynamic **MVVM UI framework** and exposed core gameplay systems to **Blueprints**, shipping it as a standalone plugin with its own **UnrealBuildTool** module and streamlining iteration for designers and technical artists.
- Implemented campaign layer gameplay systems including hero recruitment, skill trees and player state machines, and a settlement designer tool with grid snapped placement and Stronghold style wall building.

VR Gameplay Programmer, i3 Simulations, Bangalore, India

August 2022 to April 2024

- SimGym** (Unreal Engine, PC VR): built a **scenario creator** with a multigrid system, letting users author detailed simulations without engineering support, inside a fixed VR frame budget.
- ResusVR** (Unity): built the scenario designer as a **ScriptableObject** framework, letting designers author their own scenarios without code. Implemented the **analytics framework** and automated the localisation pipeline.
- ENT Surgery VR** (Unity): profiled and optimised the simulation, and implemented an **FFmpeg** based session recorder.
- Hypotensive Shock** (Unity): ported the project solo off a legacy Unity version and implemented its analytics.

Technical Interviewer and Code Reviewer, Outscal, Remote

October 2022 to August 2023

- Ran technical interviews and reviewed C++ and Unity submissions for game programming candidates, giving structured written feedback on architecture, code quality and problem decomposition.

Project Portfolio

Benji 2D, a 2D Game Engine Built from Scratch

C++, SDL2

- Built a signature based ECS with packed component pools, a frame rate independent game loop, a type indexed event bus and a Dear ImGui debug layer, wired together through a CMake build.

Automata Engine, Cellular Automata Simulator

C++20, SDL2

- Rewrote the same simulation four times against one benchmark, from the naive version through bitboards and multithreading, to measure exactly what each optimisation bought.

Cartino, Top Down 3D Bullet Hell

Unity, C#

- Android game where the player drives and the car handles its own targeting and fire, built while learning Unity and C#.

Education

APJ Abdul Kalam Technological University, B.Tech in Computer Science Engineering, Kerala, India

2022

- Full Stack Game Development, Outscal, 2020 to 2022.